INCH POUND

MIL-DTL-9395/33B
30 May 2001

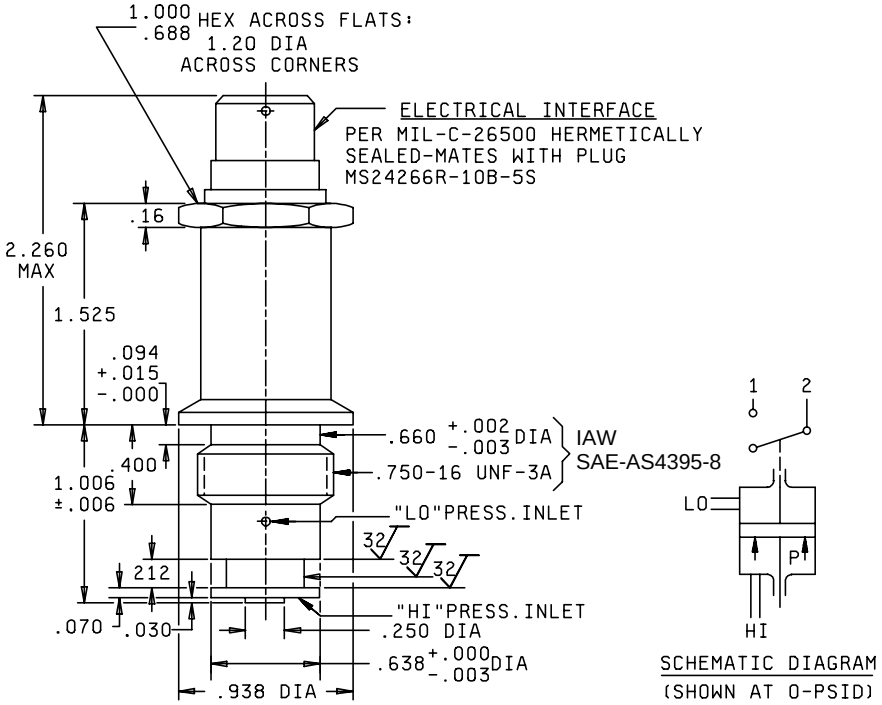
SUPERSEDING
MIL-DTL-9395/33A(USAF)
18 September 1979

DETAIL SPECIFICATION SHEET

SWITCHES, PRESSURE, (DIFFERENTIAL), TYPE III, 5 AMPERES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the pressure switches described herein shall consist of this document and the latest issue of Specification MIL-S-9395.



INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.094	2.39	.650	16.76
.002	.05	.160	4.06	.750	19.05
.003	.08	.212	5.38	.938	23.83
.006	.15	.250	6.35	1.000	25.40
.015	.38	.400	10.16	1.006	25.55
.030	.76	.533	13.54	1.120	28.45
.070	1.78	.638	16.21	1.525	38.74
				2.260	57.40

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ± .005 (.13 mm) for three place decimals and .03 (.76 mm) for two place decimals.

FIGURE 1. Switch.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Dimensions, configuration, and electrical schematic: See figure.

Weight: 0.14 pound, maximum.

Calibration: See tables I, II, III, IV, and V.

Proof pressure: 3,000 lb_f/in^2 high over low and simultaneously.

System pressure: 1,500 lb_f/in^2 high over low and simultaneously.

Burst pressure: 7,500 lb_f/in^2 high over low and simultaneously.

Electrical ratings:
Operating voltage: - 28 Vdc.
Current rating - 5 amperes resistive
3 amperes inductive.

Seal:
High pressure chamber: Media proof. Subject switches to proof pressure for 2 minutes using hydraulic fluid per MIL-H-5606 with chamber pressure continuously being monitored. Isolate the chamber at proof pressure with the chamber disconnected from the pressure source. Under that condition, the pressure shall not drop more than 1 lb_f/in^2 for the first 30 seconds to allow stabilization of test equipment; no pressure loss is allowed thereafter.
Low pressure chamber: Media proof. Tested as above except low pressure port is connected to high pressure port.
Electrical chamber: Unsealed.
Electrical connector: See figure.

Pressure ports: See figure.

Media: Hydraulic fluid per MIL-H-5606 or MIL-PRF-83282.

High temperature (operating and nonoperating): B (275°F).

Low temperature (operating and nonoperating): D (-65°F).

Altitude: C (70,000 feet).

Shock: C (100 G).

Vibration: S (test condition D, method 204 of MIL-STD-202).

Life mechanical: A (100,000 cycles).

Life electrical: C (50,000 cycles).

Acceleration: C (8 G).

Pulsation amplitude: A (0%).

Pulsation frequency: A (0 Hz).

Pressure surge: B (more than 1,000 lb_f/in^2/sec).

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PART NUMBER: The part number consists of the prefix M9395/33- followed by five code characters, the first from table I (denoting setting mode), the second from table II, the third from table III (denoting high pressure setting), the fourth from table II, and the fifth from table III (denoting the low pressure setting). The following example illustrates the method used in generating a coded part number:

EXAMPLE: M9395/33- A 2 Y Q G

Prefix

Denotes the high pressure setting is "AT or "BY" and low pressure setting is "AT" or "BY"

Denotes the coarse high pressure setting is 250 lb_f/in^2

Denotes the fine high pressure setting to be added if 0 lb_f/in^2, and the high pressure setting is max (BY)

Denotes the coarse low pressure setting is 140 lb_f/in^2

Denotes the fine low pressure setting to be added is 0, and the tolerance is $\pm$15 lb_f/in^2.

M9395/33-A2YQG is the part number for a switch that is actuated on increasing pressure by 250 lb_f/in^2 and that is deactuated on decreasing pressure at 140 $\pm$15 lb_f/in^2.

Supersession data: M9395/33-01 is superseded by M9395/33-A2YQG.

TABLE I. Code for pressure setting modes.

Code character	Setting modes	
	High pressure	Low pressure
A	AT (or BY)	AT (or BY)
B	AT (or BY)	Differential
C	Differential	AT (or BY)

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TABLE II. Codes for coarse pressure setting (within lb_f/in^2).

Code	Coarse pressure setting (lb_f/in^2)	Code	Coarse pressure setting (lb_f/in^2)	Code	Coarse pressure setting (lb_f/in^2)
A	0	N	120	1	240
B	10	P	130	2	250
C	20	Q	140	3	260
D	30	R	150	4	270
E	40	S	160	5	280
F	50	T	170	6	290
G	60	U	180	7	300
H	70	V	190	8	310
J	80	W	200	9	320
K	90	X	210	0	340
L	100	Y	220		
M	110	Z	230		

TABLE III. Code for combination of fine pressure setting and tolerance. (This quantity should be added to coarse setting selected from table II).

Fine pressure setting (lb_f/in^2)	Code characters (tolerance lb_f/in^2)											
	Min	±5	±10	±15	±20	±25	±30	±35	±40	±50	±60	Max
0	A	C	E	G	J	L	N	Q	S	U	W	Y
5	B	D	F	H	K	M	P	R	T	V	X	Z

Custodians:
Air Force - 11
DLA - CC

Review activities:
Air Force - 99

Preparing activity:
DLA - CC

(Project 5930-1730-13)

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